# EE2211 Introduction to Machine Learning

## Lecture 1

Yueming Jin
ymjin@nus.edu.sg

**Office Hour (Hybrid): Monday 11:00am - 12:00nn (Week 2-4, Week 11-13)**

© Copyright EE, NUS. All Rights Reserved.

# Course Contents

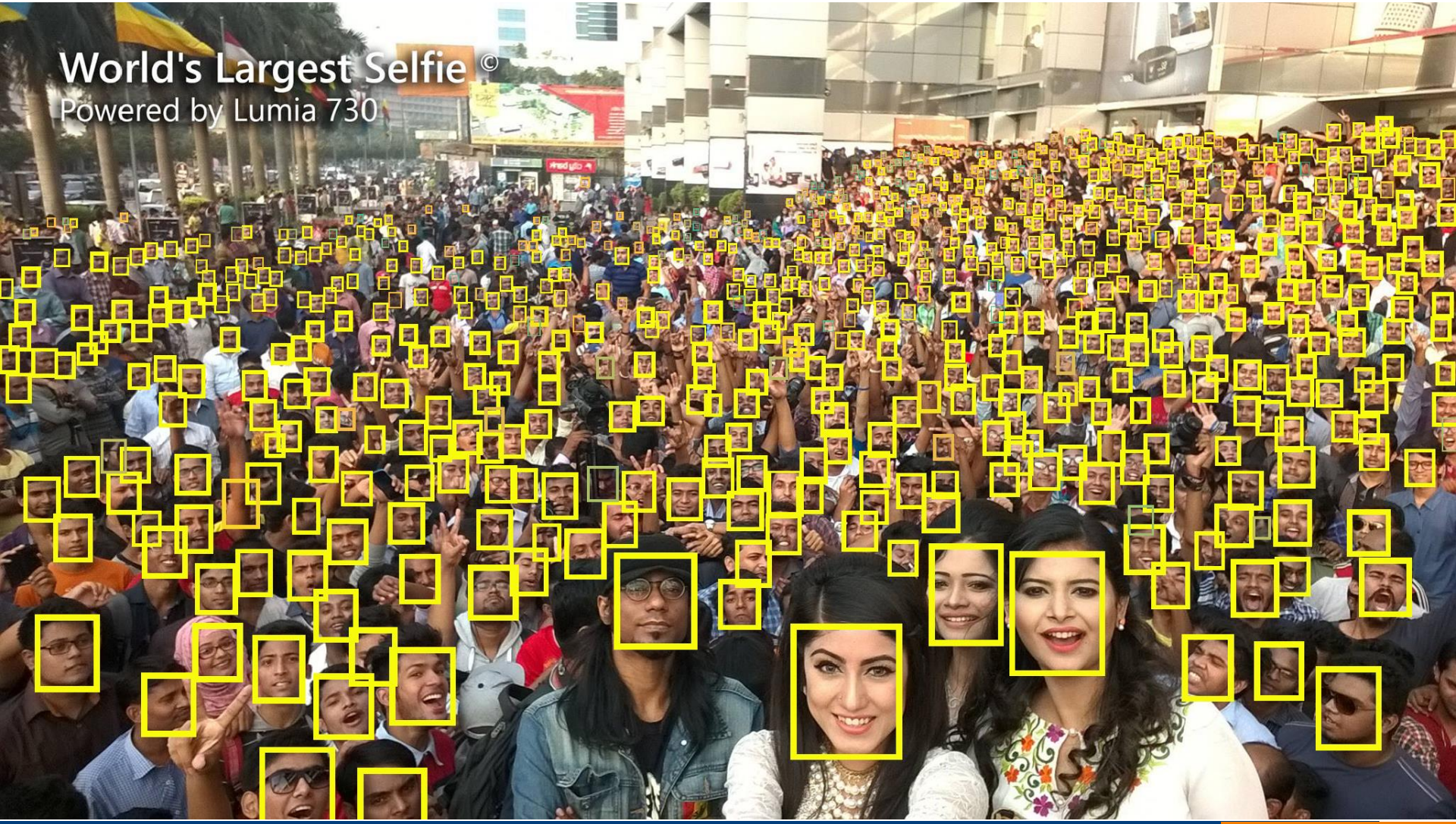
- Introduction and Preliminaries (Yueming)
  - Introduction
  - Data Engineering
  - Introduction to Probability and Statistics
- Fundamental Machine Learning Algorithms I (Shaojing)
  - Systems of linear equations
  - Least squares, Linear regression
  - Ridge regression, Polynomial regression
- Fundamental Machine Learning Algorithms II (Shaojing)
  - Over-fitting, bias/variance trade-off
  - Optimization, Gradient descent
  - Decision Trees, Random Forest
- Performance and More Algorithms (Yueming)
  - Performance Issues
  - K-means Clustering
  - Neural Networks

© Copyright EE, NUS. All Rights Reserved.

# World’s Largest Selfie

© Copyright EE, NUS. All Rights Reserved.

# World’s Largest Selfie




© Copyright EE, NUS. All Rights Reserved.

# Recent Advances

ChatGPT 3.5 Sign up

**You**

write a poem about Singapore

**ChatGPT**

In a city-state where dreams ignite,
Singapore gleams with vibrant light,
Skyscrapers soar, a modern maze,
Where history meets tomorrow's craze.

Gardens bloom in meticulous grace,
Supertrees whisper, a futuristic embrace,
Marina Bay sparkles, a jewel aglow,
Reflecting a nation's spirit, in ebb and flow.

From hawker stalls to haute cuisine,
A melting pot where cultures convene,
Peranakan hues and Tamil beats,
Chinatown's bustle, each story repeats.

Prompt: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.

© Copyright EE, NUS. All Rights Reserved.

# Outline

- What is machine learning?
  - Three Definition(s)
- When do we need machine learning?
  - Sometimes we need, sometimes we don't
- Applications of machine learning
- Types of machine learning
  - Supervised, Unsupervised, Reinforcement Learning
- Walking through a toy example on classification
- Inductive vs. Deductive Reasoning

© Copyright EE, NUS. All Rights Reserved.

# What is machine learning?

Learning is any process by which a system improves performance from experience.

- Herbert Simon

A computer program is said to learn

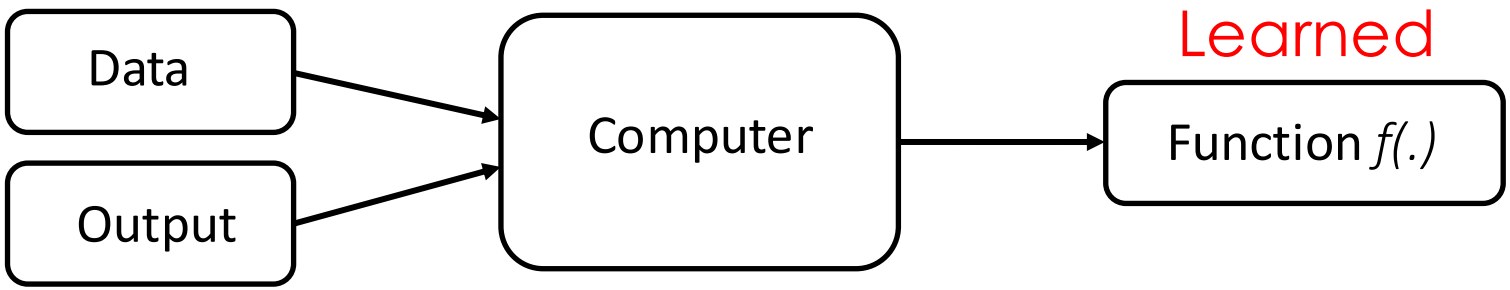
- from ***experience E***
- with respect to some class of ***tasks T***
- and performance ***measure P***,

if its performance at tasks in T, as measured by P, improves with experience E.

- Tom Mitchell

© Copyright EE, NUS. All Rights Reserved.

# Machine Learning (Supervised Learning)




**Data** **Output**

**Cat**

$f($ $) =$ 'cat'

$\longrightarrow$ $f(\,.\,)$ such that

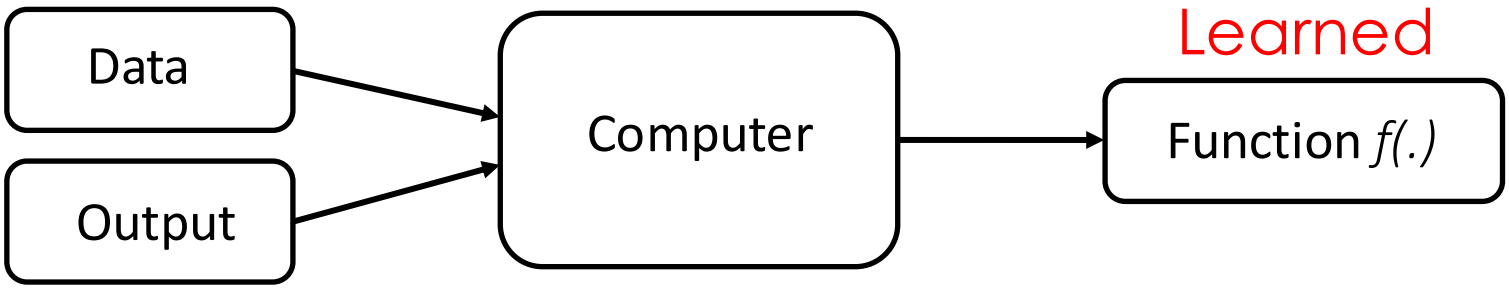
⋮

**Dog**

$f($ $) =$ 'dog'

© Copyright EE, NUS. All Rights Reserved.

# Machine Learning (Supervised Learning)




**Data** **Output**

Cat

⋮

Dog

$\longrightarrow f(\,.\,)$

**When applied**

$f($ $) \longrightarrow$ Cat !

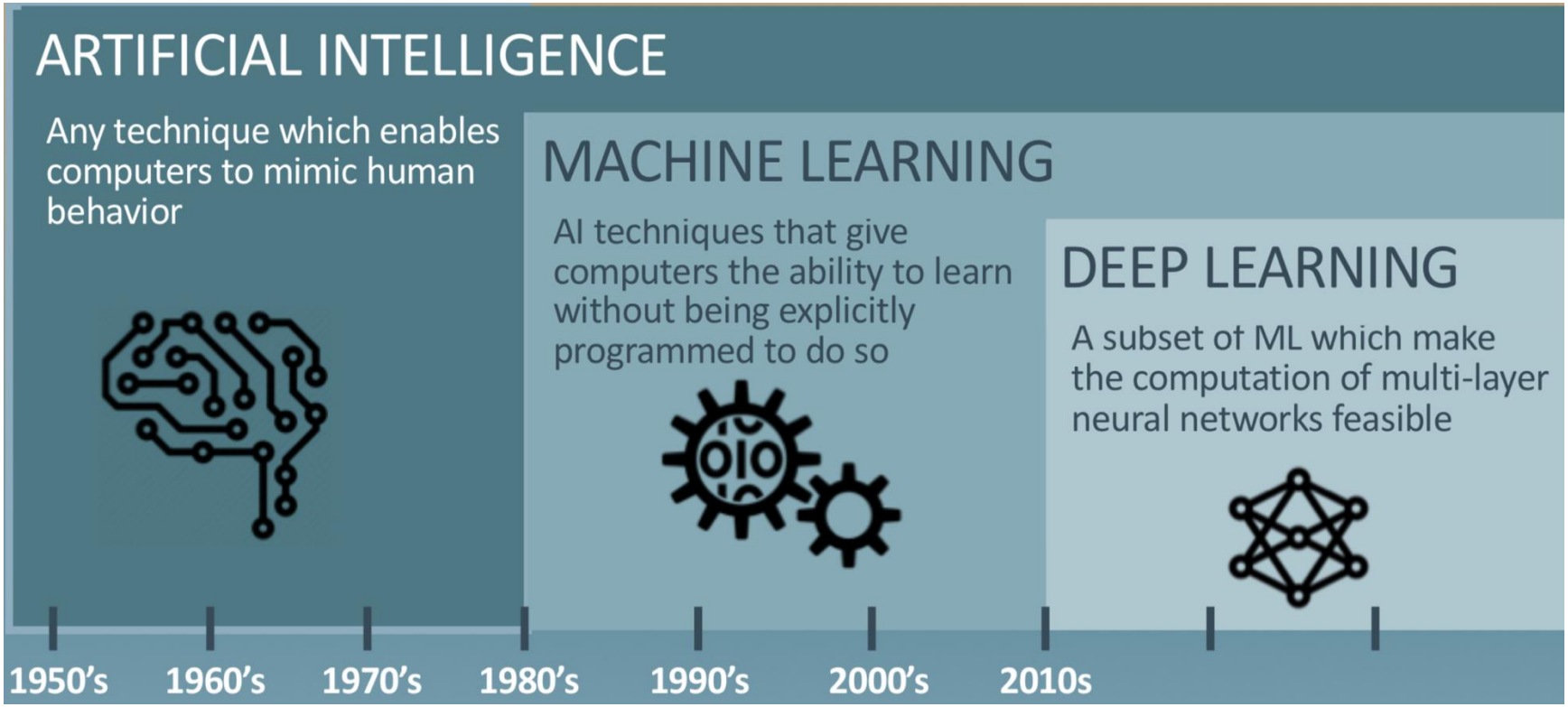
**New image**

Machine Learning: field of study that gives computers the ability to learn without being explicitly programmed

- Arthur Samuel

© Copyright EE, NUS. All Rights Reserved.

# AI, Machine Learning, and Deep Learning




**Example of AI but not ML: Deductive Reasoning**

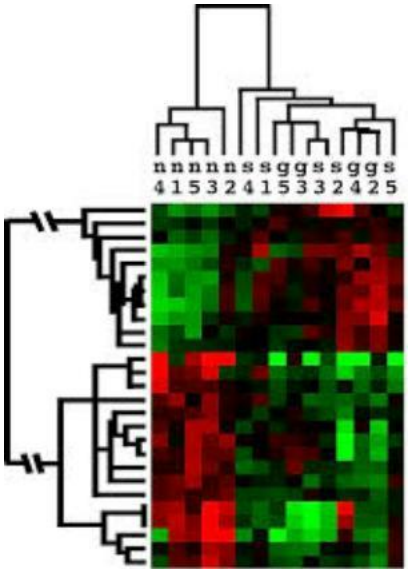
NUS is in Singapore, Singapore is in Asia -> NUS is in Asia

© Copyright EE, NUS. All Rights Reserved.

# When do we need machine learning?

Lack of human expertise
(Navigating on Mars)

Involves huge amount of data
(Genomics)




**Learning is not always useful:**

No need to “learn” to calculate payroll!

My Salary = Days_of_work * Daily Salary + Bonus

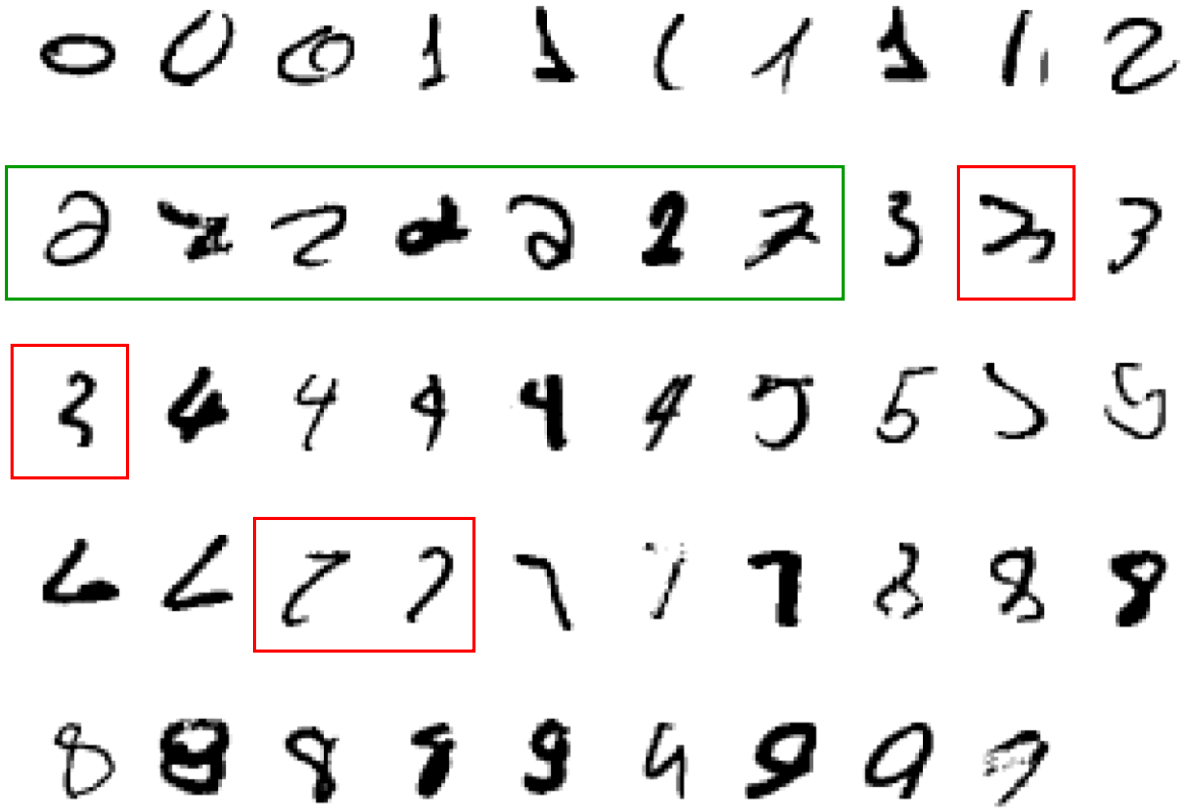
© Copyright EE, NUS. All Rights Reserved.

# Application of Machine Learning

Task T, Performance P, Experience E

T: Digit Recognition
P: Classification Accuracy
E: Labelled Images

Labels -> Supervision!



© Copyright EE, NUS. All Rights Reserved.

# Application of Machine Learning

Task T, Performance P, Experience E

T: Email Categorization
P: Classification Accuracy
E: Email Data, Some Labelled

© Copyright EE, NUS. All Rights Reserved.

# Application of Machine Learning

Task T, Performance P, Experience E

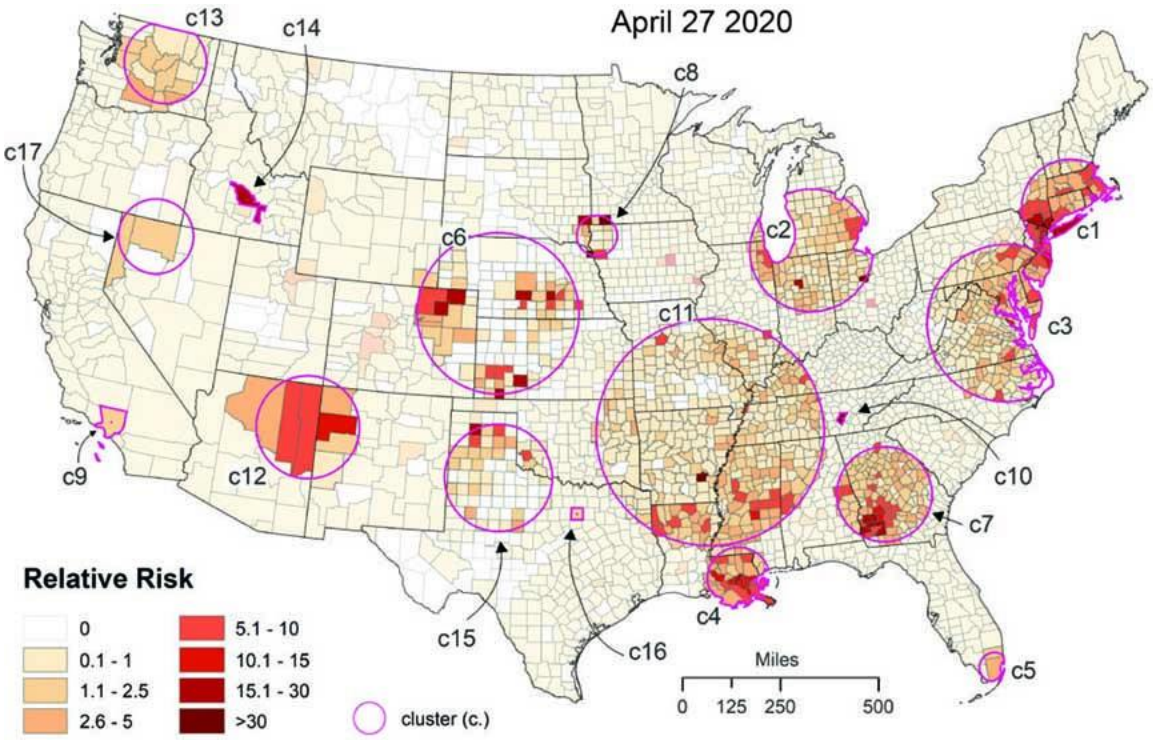
T: Playing Go Game
P: Chances of Winning
E: Records of Past Games

© Copyright EE, NUS. All Rights Reserved.

# Application of Machine Learning

Task T, Performance P, Experience E

T: Identifying Covid-19 Clusters
P: Small Internal Distances
Larger External Distances
E: Records of Patients




© Copyright EE, NUS. All Rights Reserved.

*Web Search Engine*

*Product Recommendation*

*Language Translation*

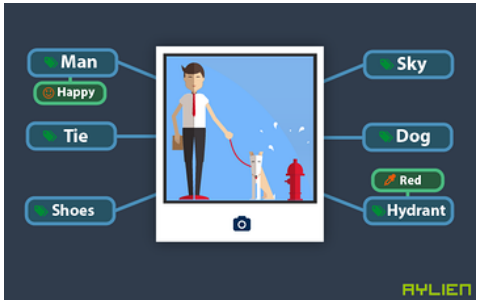
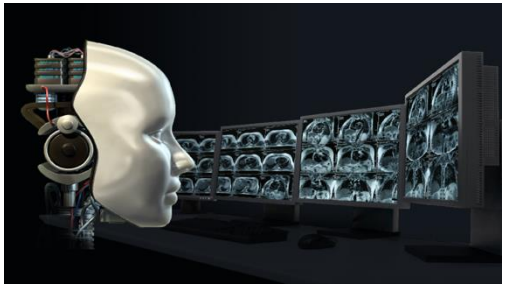



*Photo Tagging*

*Virtual Personal Assistant*

*Portfolio Management*

*Traffic Prediction*



*Medical Diagnosis*

*Algorithmic Trading*

© Copyright EE, NUS. All Rights Reserved.

# Types of Machine Learning

### Supervised Learning

*Input:*
1) Training Samples,
2) Desired Output (Teacher/Supervision)

*Output:*
A rule that maps input to output

### Unsupervised Learning

*Input:*
Samples

*Output:*
Underlying patterns in data

### Reinforcement Learning

*Input:*
Sequence of States, Actions, and Delayed Rewards

*Output:*
Action Strategy: a rule that maps the environment to action

© Copyright EE, NUS. All Rights Reserved.

# Types of Machine Learning

**Supervised Learning**

*Input:*
1) Training Samples,
2) Desired Output (Teacher/Supervision)

*Output:*
A rule that maps input to output

**Unsupervised Learning**

*Input:*
Samples

*Output:*
Underlying patterns in data

**Reinforcement Learning**

*Input:*
Sequence of States, Actions, and Delayed Rewards

*Output:*
Action Strategy: a rule that maps the environment to action

**Data** **Output**

**Cat**

⋮

**Dog**

$\longrightarrow$ $f(.)$ such that

$f($ $)$ = 'cat'

$f($ $)$ = 'dog'

© Copyright EE, NUS. All Rights Reserved.

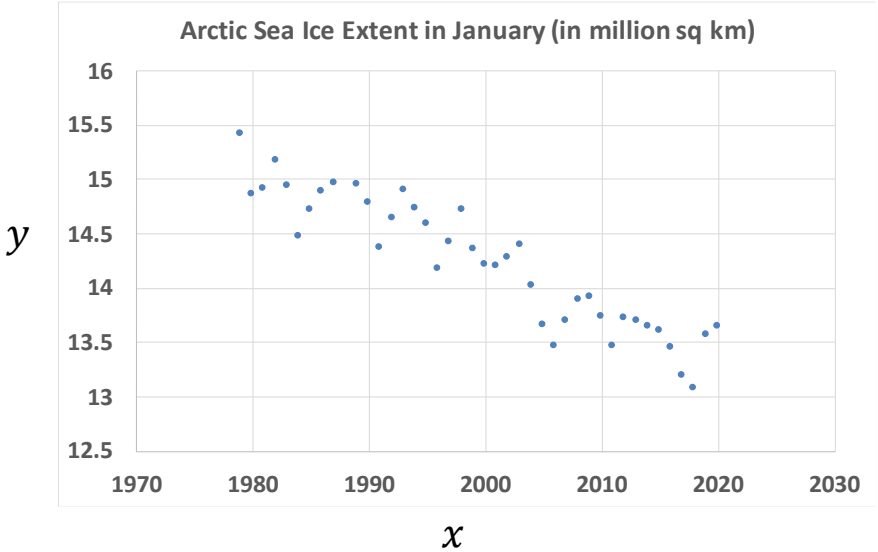
# Supervised Learning

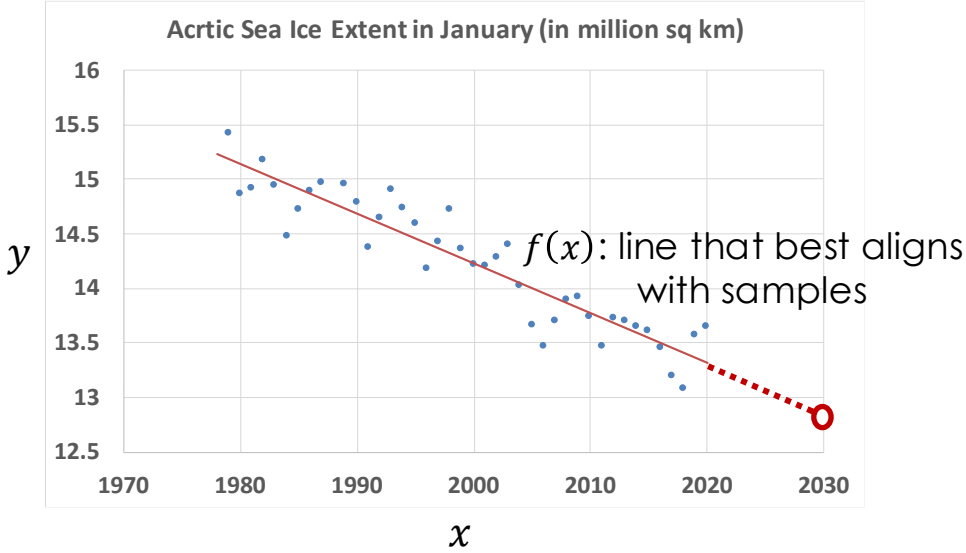
$x$ → **Regression** → $y$ (Continuous)

$x$ → **Classification** → $y$ (Categorical)

**Regression**

- Given $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)$
- Learn a function $f(\mathbf{x})$ to predict real-valued $y$ given $\mathbf{x}$




© Copyright EE, NUS. All Rights Reserved.

# Supervised Learning

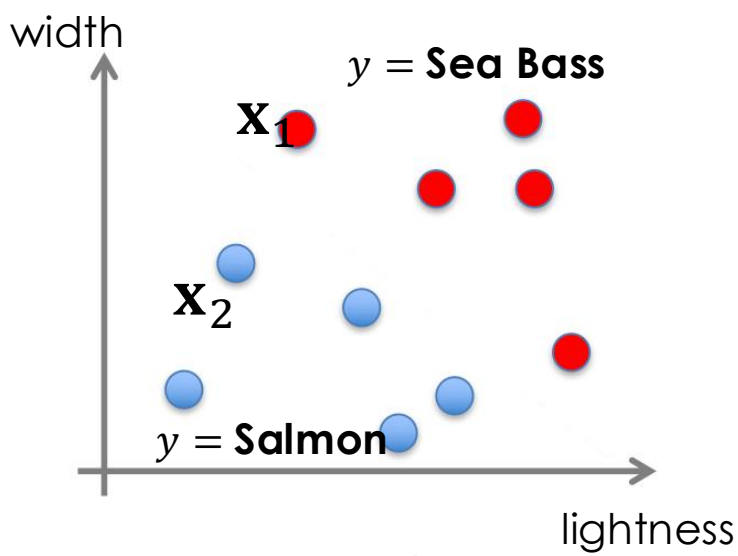
$x$ → **Regression** → $y$ (Continuous)

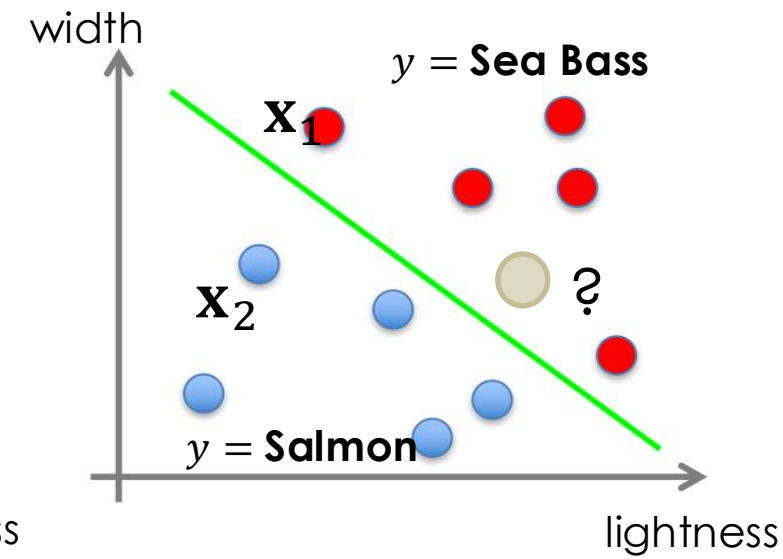
$x$ → **Classification** → $y$ (Categorical)

**Classification**

- Given $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)$
- Learn a function $f(\mathbf{x})$ to predict categorical $y$ given $\mathbf{x}$

**Feature Space**




$f(x)$: line that separates two classes

© Copyright EE, NUS. All Rights Reserved.

# Types of Machine Learning

**Supervised Learning**

*Input:*
1) Training Samples,
2) Desired Output (Teacher/Supervision)

*Output:*
A rule that maps input to output

**Unsupervised Learning**

*Input:*
Samples

*Output:*
Underlying patterns in data

**Reinforcement Learning**

*Input:*
Sequence of States, Actions, and Delayed Rewards

*Output:*
Action Strategy: a rule that maps the environment to action

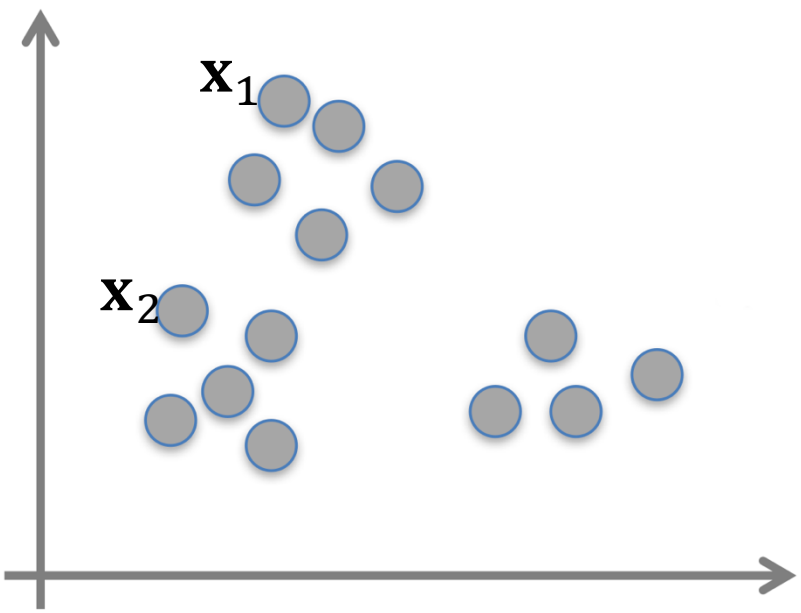
**Key different w.r.t. supervised learning**
**No Label/Supervision is given!**

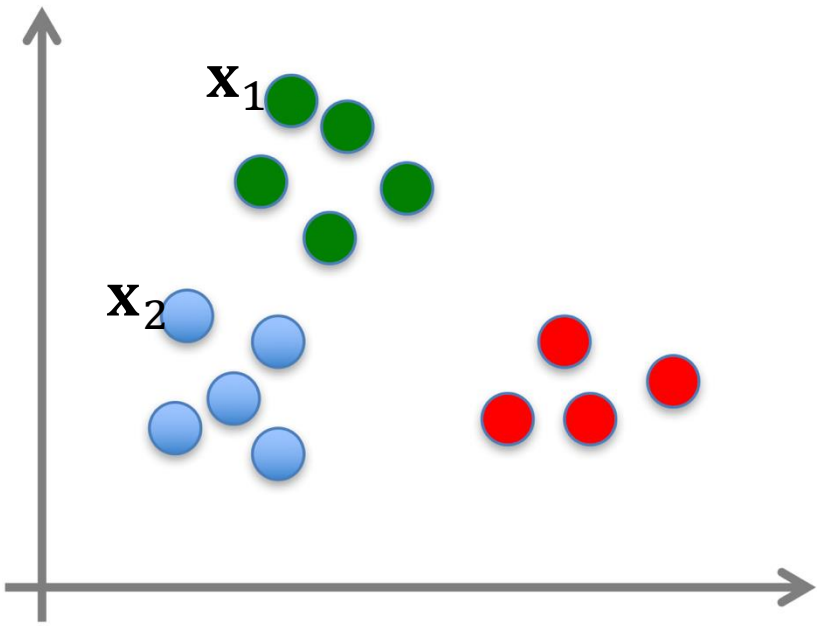
© Copyright EE, NUS. All Rights Reserved.

# Unsupervised Learning

**Clustering**

- Given $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N$, without labels
- Output Hidden Structure Behind




**No Label/Supervision is given!**

© Copyright EE, NUS. All Rights Reserved.

# Types of Machine Learning

**Supervised Learning**

*Input:*
1) Training Samples,
2) Desired Output (Teacher/Supervision)

*Output:*
A rule that maps input to output

**Unsupervised Learning**

*Input:*
Samples

*Output:*
Underlying patterns in data

**Reinforcement Learning**

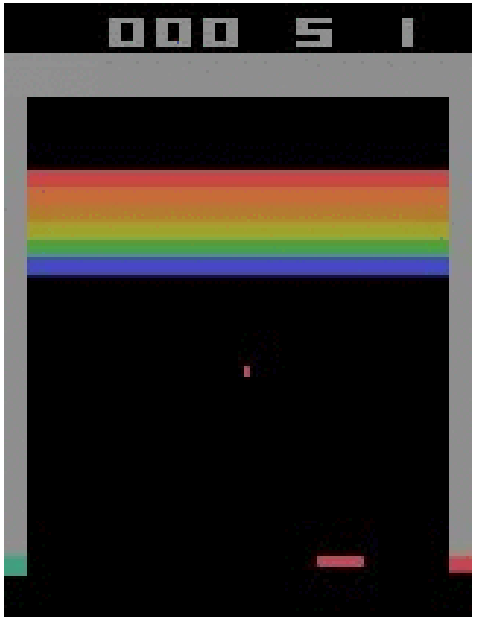
*Input:*
Sequence of States, Actions, and Delayed Rewards

*Output:*
Action Strategy: a rule that maps the environment to action

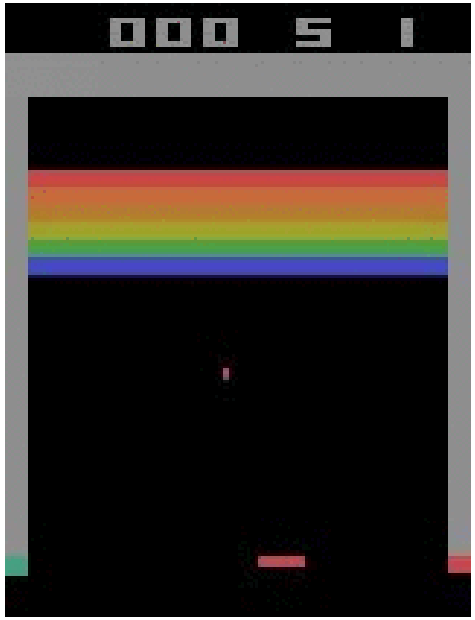
© Copyright EE, NUS. All Rights Reserved.

# Reinforcement Learning

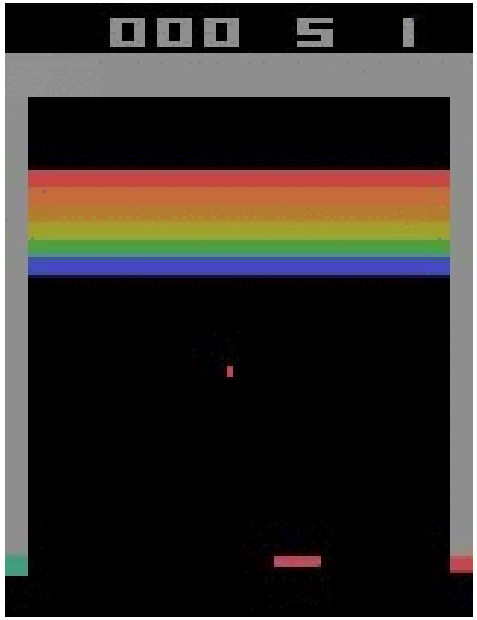
## Breakout Game




Initial Performance




Training 15 minutes

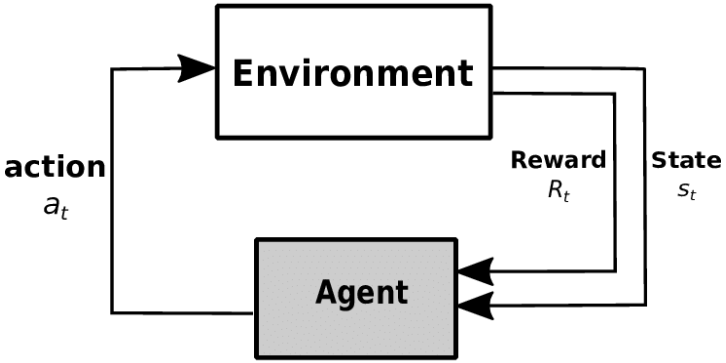


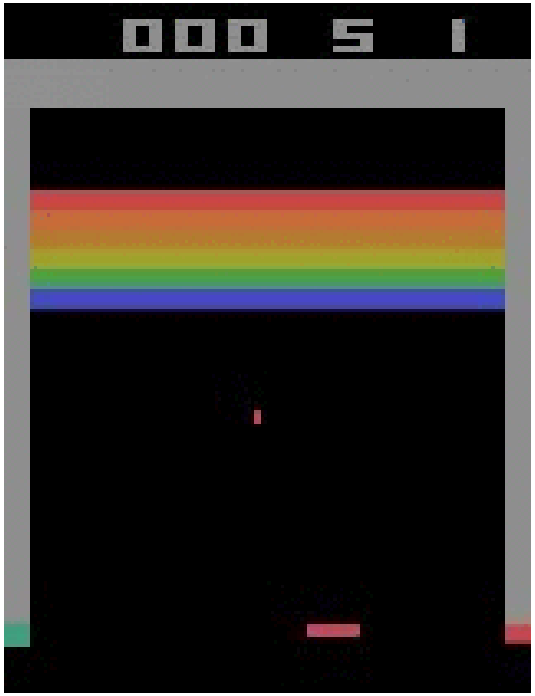

Training 30 minutes

© Copyright EE, NUS. All Rights Reserved.

# Reinforcement Learning

- Given sequence of states $\boldsymbol{S}$ and actions $\boldsymbol{A}$ with (delayed) rewards $\boldsymbol{R}$
- Output a policy $\pi(a, s)$, to guide us what action $a$ to take in state $s$




$\boldsymbol{S}$: Ball Location, Paddle Location, Bricks

$\boldsymbol{A}$: left, right

$\boldsymbol{R}$:

<u>*positive reward*</u>
Knocking a brick, clearing all bricks

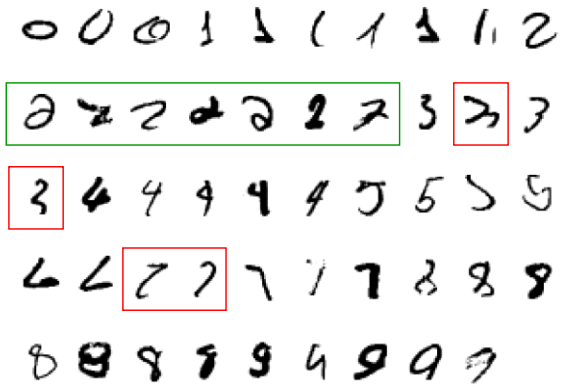
<u>*negative reward*</u>
Missing the ball

<u>*zero reward*</u>
Cases in between

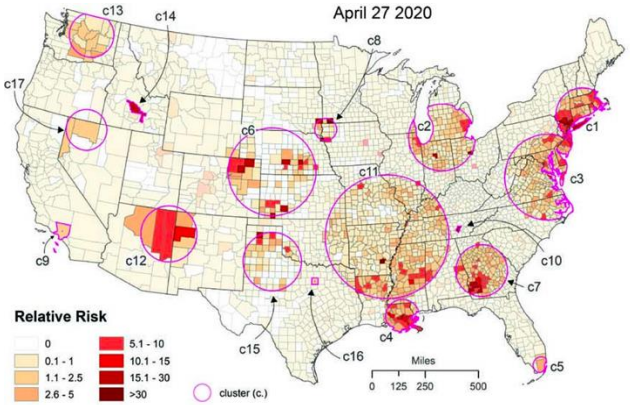
© Copyright EE, NUS. All Rights Reserved.

Supervised
Unsupervised
Reinforcement

# Quiz Time!



Supervised



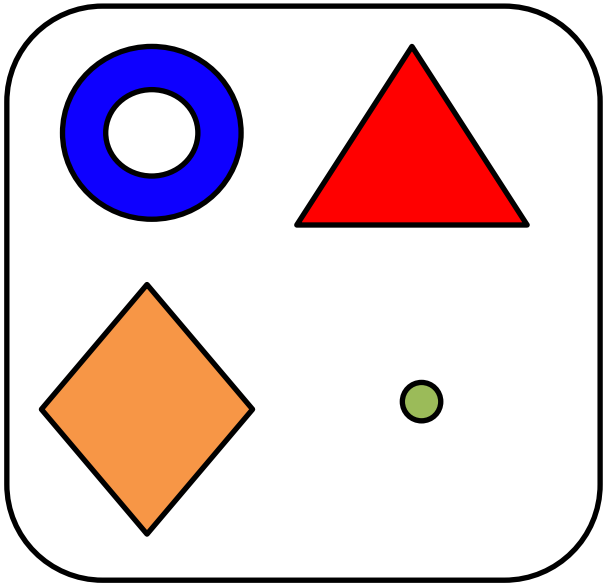

Unsupervised

Supervised

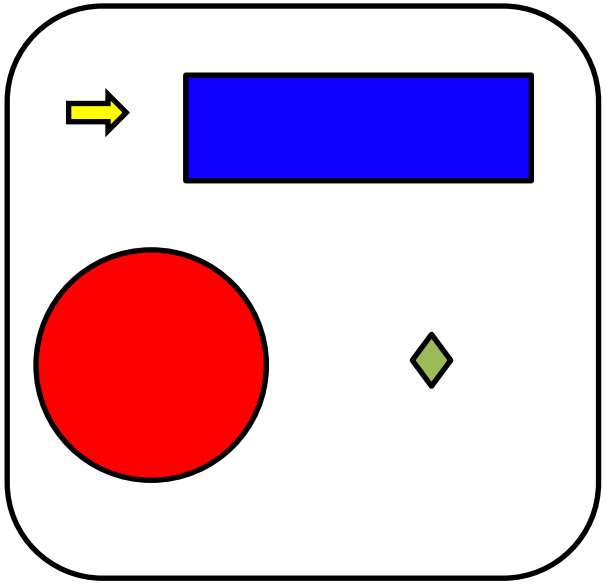
Reinforcement

© Copyright EE, NUS. All Rights Reserved.

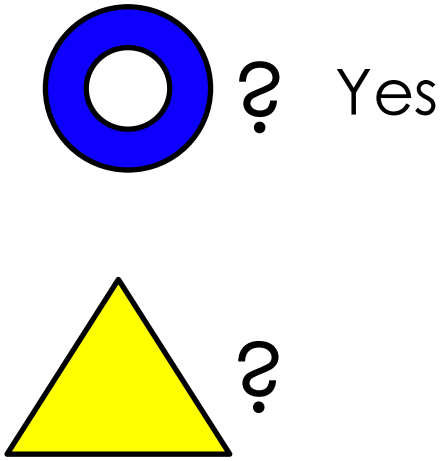
# Walking Through A Toy Example: Token Classification



Yes



No



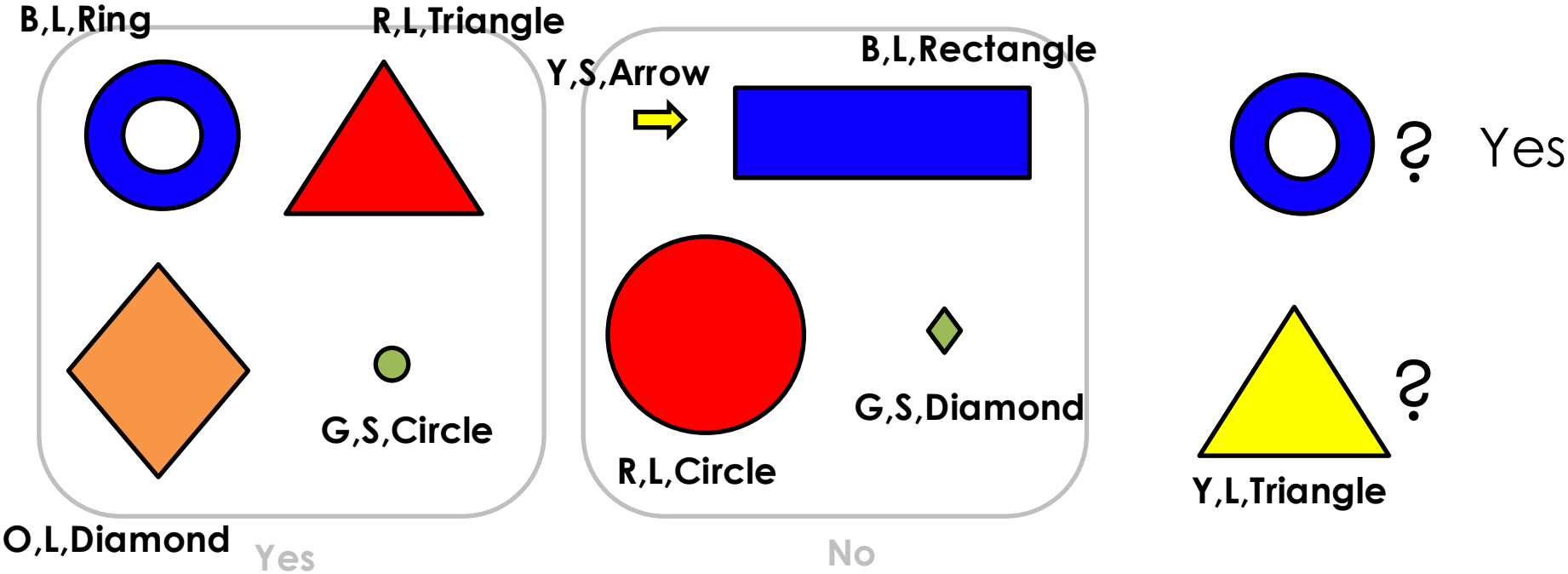

***Step1: Feature Extraction***
Extract Attributes of Samples

→

***Step2: Sample Classification***
Decide Label for a Sample

© Copyright EE, NUS. All Rights Reserved.

# Walking Through A Toy Example: Token Classification




***Step 1: Feature Extraction***

*Color, Size, Shape*

© Copyright EE, NUS. All Rights Reserved.

# Walking Through A Toy Example: Token Classification

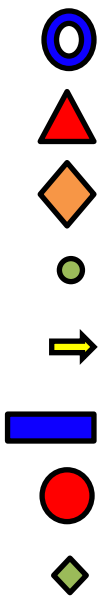
## Feature Extraction

| | Color | Size | Shape | Label |
|---|---|---|---|---|
| | Blue | Large | Ring | Yes |
| | Red | Large | Triangle | Yes |
| | Orange | Large | Diamond | Yes |
| | Green | Small | Circle | Yes |
| | Yellow | Small | Arrow | No |
| | Blue | Large | Rectangle | No |
| | Red | Large | Circle | No |
| | Green | Small | Diamond | No |
| | Yellow | Large | Triangle | ? |

© Copyright EE, NUS. All Rights Reserved.

# Walking Through A Toy Example: Token Classification

## Feature Extraction



| Color | Size | Shape | Label |
|---|---|---|---|
| Blue | Large | Ring | Yes |
| Red | Large | Triangle | Yes |
| Orange | Large | Diamond | Yes |
| Green | Small | Circle | Yes |
| Yellow | Small | Arrow | No |
| Blue | Large | Rectangle | No |
| Red | Large | Circle | No |
| Green | Small | Diamond | No |

© Copyright EE, NUS. All Rights Reserved.

# Walking Through A Toy Example: Token Classification

## Feature Extraction

| Color | Size | Shape | Label |
|---|---|---|---|
| Blue | Large | Ring | Yes |
| Red | Large | Triangle | Yes |
| Orange | Large | Diamond | Yes |
| Green | Small | Circle | Yes |
| Yellow | Small | Arrow | No |
| Blue | Large | Rectangle | No |
| Red | Large | Circle | No |
| Green | Small | Diamond | No |

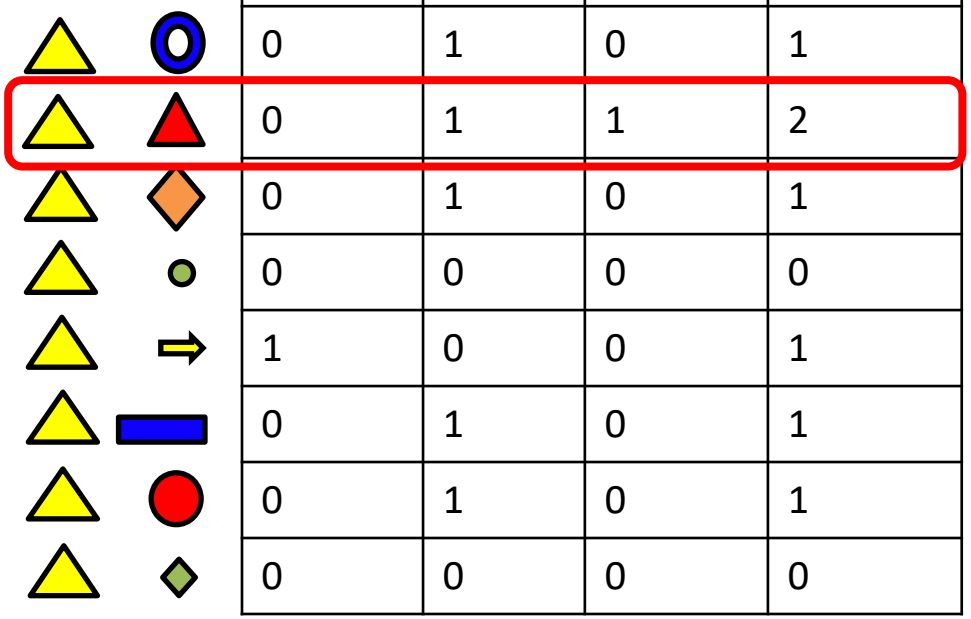
## Similarity

| Color | Size | Shape | Total |
|---|---|---|---|
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 2 |
| 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 |

© Copyright EE, NUS. All Rights Reserved.

# Walking Through A Toy Example: Token Classification

## Similarity



| Color | Size | Shape | Total |
|---|---|---|---|
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 2 |
| 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 |

**Nearest Neighbor Classifier:**

1) Find the "nearest neighbor" of a sample in the feature space

2) Assign the label of the nearest neighbor to the sample

© Copyright EE, NUS. All Rights Reserved.

# Inductive vs. Deductive Reasoning

- Main Task of Machine Learning: to make inference

**Two Types of Inference**

**Inductive**

- To reach **probable** conclusions
- Not all needed information is available, causing uncertainty

**Deductive**

- To reach logical conclusions **deterministically**
- All information that can lead to the correct conclusion is available

Probability and Statistics

Rule-based reasoning

NUS is in Singapore, Singapore is in Asia -> NUS is in Asia

© Copyright EE, NUS. All Rights Reserved.

# Inductive Reasoning

Note: humans use inductive reasoning all the time and not in a formal way like using probability/statistics.

Ref: Gardener, Martin (March 1979). "MATHEMATICAL GAMES: On the fabric of inductive logic, and some probability paradoxes" (PDF). *Scientific American*. **234**

© Copyright EE, NUS. All Rights Reserved.

# Summary by Quick Quiz

**Three Components in ML Definition**

Task T, Performance P, Experience E

**Two Types of Supervised Learning**

Classification, Regression

**Three Types of in ML**

Supervised Learning
Unsupervised Learning
Reinforcement Learning

**One Type of Unsupervised Learning**

Clustering

**Inductive and Deductive**

Inductive: Probable
Deductive: Rule-based

**Example of a Classifier Model**

Nearest Neighbor Classifier

© Copyright EE, NUS. All Rights Reserved.

# Practice Question
## (Type of Question to Expect in Exams)

Which of the following statement is true?

A. Nearest Neighbor Classifier is an example of unsupervised learning
B. Nearest Neighbor Classifier is an example of deductive learning
C. Nearest Neighbor Classifier is an example of feature selection
D. None of the above is correct.

© Copyright EE, NUS. All Rights Reserved.

© Copyright EE, NUS. All Rights Reserved.